Scalability & Future Plan

Date	07-07-2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No soil type input — model relies only on N, P, K, and climate values	Recommendations don't account for soil texture (sandy, loamy, clay, black), which affects real-world suitability	Medium
2	Free-tier hosting cold start	First request after inactivity takes 30-50 seconds while the Render server restarts	Low
3	Static training dataset — no live weather integration	Recommendations are based on historical averages, not real-time local weather conditions	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	**User Load**	Single Render free-tier instance, no auto-scaling, spins down when idle	Upgrade to a paid Render tier with always-on hosting and auto-scaling if concurrent usage grows
2	**Data Storage**	Static CSV dataset and .pkl model files bundled directly in the repository	Move dataset and model artifacts to a managed database or object storage (e.g. PostgreSQL, AWS S3) if the dataset grows or needs frequent updates
3	**Performance**	Model and scaler loaded once at startup; single-threaded Flask dev server locally, gunicorn in production	Increase gunicorn worker count for higher concurrent request handling; consider caching repeated predictions

4	**Security**	No authentication or rate limiting; relies on Render's default HTTPS	Add basic input sanitization, request rate limiting, and HTTPS enforcement if the app is opened to public/production traffic at scale

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add soil type as an input feature (requires a richer dataset and model retraining), plus basic fertilizer recommendations	2-3 months	More personalized, soil-aware crop recommendations
Phase 3	Integrate a live weather API to auto-fill temperature, humidity, and rainfall based on user location	3-6 months	Recommendations reflect real-time conditions instead of historical averages
Phase 4	Add regional language support and a mobile-friendly progressive web app (PWA)	6-12 months	Wider accessibility for non-English-speaking and mobile-first farmers